



Exercises session 3

PTIME

1 Definitions

Definition 1.1 Let $t : \mathbb{N} \rightarrow \mathbb{N}$ be a function. We define the (deterministic) **time complexity class** $DTIME(t(n))$ as:

$$DTIME(t(n)) = \{L \mid L \text{ is decided by an } O(t(n)) \text{ time deterministic Turing machine}\}.$$

Definition 1.2 $PTIME$ (or $\mathbf{P}$) is the class of all languages decidable in deterministic polynomial time on a single-tape TM

$$PTIME = \bigcup_k DTIME(n^k).$$

Definition 1.3 (Vertex colouring) A vertex colouring of a graph $G = (V, E)$ with k colours is a function

$$c : V(G) \rightarrow \{1, \dots, k\}$$

such that adjacent nodes have different colours, that is

$$(u, v) \in E \rightarrow c(u) \neq c(v)$$

2 Session exercises

1. Prove that $\mathbf{P}$ is closed under intersection, union, concatenation and complement.
2. Show that the following language

$$\text{TRIANGLE} = \{\langle G \rangle : G \text{ has a triangle}\}$$

is in $\mathbf{P}$

3. Consider the following problem:

$$2\text{-COL} = \{G : G \text{ has a 2-colouring}\}$$

- Design an algorithm that decides if a graph has a 2-colouring

- Relate this problem to 2-SAT and conclude that 2-COL is in PTIME.
4. We saw in classes that Horn-SAT is solvable in polynomial time. However, sometimes not all clauses can be satisfied. In such cases, we seek an assignment that satisfies the maximum number of clauses—this is the **Max-Horn-SAT** problem.
- Given the following Horn formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (\neg x_3) \wedge (x_4)$$

- (a) Determine if a satisfying assignment exists.
- (b) If the formula is unsatisfiable, determine the **maximum number of clauses** that can be satisfied.
- (c) Provide an example of an assignment that **maximizes satisfied clauses**.
- (d) Can you think of an efficient algorithm for solving Max-Horn-SAT?